Iliass Sijelmassi

Paris, France | +33 6 48 74 04 04 | iliass.sijelmassi-idrissi@polytechnique.edu | iliasssijelmassi.com | github.com/iliassSjm

Summary

Master in Data Science & AI candidate (École Polytechnique & HEC Paris, GPA 3.8/4.0) and Stanford visiting AI researcher. Experience developing machine learning models for clinical prediction using large-scale physiological waveform and healthcare datasets. Strong background in statistics, predictive modeling, optimization, and data-driven research. Seeking Quantitative Researcher roles.

Education

École Polytechnique — MSc Data Science & AI (Year 1) GPA: 3.8/4.0 *Sep 2024 – Jun 2025*
- **Coursework:** Probability & Statistics, Linear Algebra & Regression, Optimization, Deep Learning, Machine Learning I & II.

HEC Paris — MSc Data Science & AI (Year 2) *Sep 2025 – Jun 2026 (expected)*
- **Relevant coursework:** Machine Learning for Financial Markets, Time Series, Causal Inference, Reinforcement Learning, Interpretability & Algorithmic Fairness, ML Ops & Cloud Engineering.

INP–ENSEEIHT, Toulouse — Dipl. Eng. in CS, Telecom & Applied Math *Sep 2020 – Sep 2023*
- **Coursework:** Probability, Statistics, Optimization, Reinforcement Learning, Algorithms & Data Structures, High-Performance Computing.

Experience

Visiting Student Researcher | ***Stanford University**, Stanford, CA* *Mar 2026 – Present*
- Building cardiac-event prediction models under Dr. Louise Sun on a **1.5 TB** longitudinal clinical dataset (ECG, echocardiogram, and arterial-pressure waveforms); **first-author paper in preparation**.
- Designed a leakage-controlled, multi-horizon evaluation benchmarking **XGBoost / Random Forest** and deep models against clinical risk scores, with strict time-ordered splits, informative-missingness encoding (**MIA**), and **FFT**-based waveform features.

Data Scientist Intern | ***Crédit Agricole CIB (CACIB)**, Paris* *Mar 2025 – Sep 2025*
- Modeled whether internal audit recommendations would be implemented on time across **200k+** historical records, reaching **F1** $\approx$ **0.75** and improving **ROC-AUC** over the heuristic baseline.
- Designed a dual-model approach: **regularized Logistic Regression** for coefficient-level interpretability required by auditors, alongside **XGBoost** for predictive performance.
- Drove feature selection with domain experts and validated robustness against **overfitting on out-of-time data**; built reproducible pipelines in **Python**, **Dataiku**, and **MLflow**.

Software Engineering Intern | ***Infosys – Renault**, Paris* *Mar 2023 – Oct 2023*
- Built backend modules for supply-chain platforms in **Java** and **PostgreSQL**.

Quantitative Research & Engineering Projects

Crypto Perpetuals Alpha Strategy | *ML for Financial Markets, HEC Paris* *2026*
- Ranked **#1 of 13 teams (~50 students)** on out-of-sample Sharpe for a dollar-neutral long-short strategy across **411 cryptocurrency perpetual futures** (gross of transaction costs).
- Engineered ~500 features filtered via **Spearman IC** and correlation pruning into a **Ridge + XGBoost** ensemble, under strict expanding-window **walk-forward** with per-window refitting of IC filter, PCA, scaler, and models to prevent look-ahead bias.

Leukemia Survival Prediction — QRT / ENS Data Challenge | *Cox PH, Python* *2025*
- Predicted overall survival on clinical + genomic data from 24 hospitals (**4,500+ patients**) using a **Cox Proportional Hazards** model; finished **top 12%**.

Cognitive Alpha — Spatial Decision Analytics, FIFA World Cup 2022 | *Personal Project, Python* *2025–2026*
- Quantified, for each open-play pass, the gap between the player's decision and the spatially optimal alternative: trained an **Expected Threat (xT)** model via **Markov-chain value iteration** on **234k events**, with a NumPy-vectorised pitch-control engine and a **Streamlit** dashboard over $\sim$5.4 GB of tracking data.

Technical Skills & Activities

Programming: **Python** (primary), SQL (PostgreSQL), R, Bash; C++ / Java (working knowledge)
Quantitative & ML: Time-Series, Probability & Statistics, Machine Learning, Deep Learning, Optimization
Tools: PyTorch, XGBoost, Scikit-learn, NumPy, Pandas, MLflow, Docker, Git, Kafka, FastAPI, Dataiku, Linux
Self-study & Certs: Stochastic calculus & options theory (Hull, Natenberg, Wilmott); Akuna Capital Options 101 & 202; **Jane Street** puzzles
Languages: French (Native), English (Fluent), Dutch (B2), Spanish, Portuguese, Arabic
Activities: Private mathematics tutor (3+ years); **Chess** (1700 Elo); Built and sold a 200k-follower X/Twitter account to Polymarket